```
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W7D3_LlamaIndex> python w7d3_llamaindex.py

======================================================================
W7D3 - LLAMAINDEX DOCUMENT INDEXING & QUERYING
======================================================================
Documents directory: C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W7D3_LlamaIndex\documents
Ollama embedding model: nomic-embed-text:latest
Ollama LLM model: llama3.2:3b
Ollama URL: http://localhost:11434
Top-K retrieval: 3

Text documents found: 5
  - artificial_intelligence.txt
  - cloud_computing.txt
  - Document 1.txt
  - software_design.txt
  - software_engineering.txt

======================================================================
PREPARING TEXT DOCUMENTS
======================================================================

TXT files found: 5
  artificial_intelligence.txt: 205,376 bytes -> 50,000 characters used
  cloud_computing.txt: 18,475 bytes -> 13,007 characters used
  Document 1.txt: 235,162 bytes -> 50,000 characters used
  software_design.txt: 549,664 bytes -> 50,000 characters used
  software_engineering.txt: 6,015,672 bytes -> 50,000 characters used

Working directory: C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W7D3_LlamaIndex\rag_documents

======================================================================
CONFIGURING LOCAL OLLAMA
======================================================================
Ollama embedding model configured successfully.
Embedding model: nomic-embed-text:latest
```

```
Ollama LLM configured successfully.
LLM model: llama3.2:3b
SentenceSplitter configured successfully.
Chunk size: 512
Chunk overlap: 50

Local LlamaIndex configuration completed.

======================================================================
LOADING TEXT DOCUMENTS
======================================================================
Documents loaded: 6

======================================================================
PART 1 - LLAMAINDEX VECTORSTOREINDEX
======================================================================
Creating VectorStoreIndex...
Embedding documents using Ollama...

VectorStoreIndex created successfully.
QueryEngine created successfully.
Similarity top-k: 3
LLM: llama3.2:3b

======================================================================
PART 1 - LLAMAINDEX QUERYENGINE - 10 QUERIES
======================================================================

----------------------------------------------------------------------
Query 1/10
Question: What is software engineering?

ERROR executing query: llama-server reported out-of-memory during startup: ggml_backend_cpu_buffer_type_alloc_buffer: failed to alloc
ate buffer of size 15032385536
alloc_tensor_range: failed to allocate CPU buffer of size 15032385536
llama_init_from_model: failed to initialize the context: failed to allocate buffer for kv cache (status code: 500)
```

```
----------------------------------------------------------------------
Query 2/10
Question: What is software design?

ERROR executing query: llama-server reported out-of-memory during startup: ggml_backend_cpu_buffer_type_alloc_buffer: failed to alloc
ate buffer of size 15032385536
alloc_tensor_range: failed to allocate CPU buffer of size 15032385536
llama_init_from_model: failed to initialize the context: failed to allocate buffer for kv cache (status code: 500)

----------------------------------------------------------------------
Query 3/10
Question: What is artificial intelligence?

ERROR executing query: llama-server reported out-of-memory during startup: ggml_backend_cpu_buffer_type_alloc_buffer: failed to alloc
ate buffer of size 15032385536
alloc_tensor_range: failed to allocate CPU buffer of size 15032385536
llama_init_from_model: failed to initialize the context: failed to allocate buffer for kv cache (status code: 500)

----------------------------------------------------------------------
Query 4/10
Question: What is machine learning?

ERROR executing query: llama-server reported out-of-memory during startup: ggml_backend_cpu_buffer_type_alloc_buffer: failed to alloc
ate buffer of size 15032385536
alloc_tensor_range: failed to allocate CPU buffer of size 15032385536
llama_init_from_model: failed to initialize the context: failed to allocate buffer for kv cache (status code: 500)

----------------------------------------------------------------------
Query 5/10
Question: What are neural networks?

ERROR executing query: llama-server reported out-of-memory during startup: ggml_backend_cpu_buffer_type_alloc_buffer: failed to alloc
ate buffer of size 838909984
ggml_gallocr_reserve_n_impl: failed to allocate CPU buffer of size 838909984
graph_reserve: failed to allocate compute buffers
llama_init_from_model: failed to initialize the context: failed to allocate compute pp buffers (status code: 500)
```

```
----------------------------------------------------------------------
Query 6/10
Question: What is natural language processing?

ERROR executing query: llama-server reported out-of-memory during startup: ggml_backend_cpu_buffer_type_alloc_buffer: failed to alloc
ate buffer of size 15032385536
alloc_tensor_range: failed to allocate CPU buffer of size 15032385536
llama_init_from_model: failed to initialize the context: failed to allocate buffer for kv cache (status code: 500)

----------------------------------------------------------------------
Query 7/10
Question: What is cloud computing?

ERROR executing query: llama-server reported out-of-memory during startup: ggml_backend_cpu_buffer_type_alloc_buffer: failed to alloc
ate buffer of size 15032385536
alloc_tensor_range: failed to allocate CPU buffer of size 15032385536
llama_init_from_model: failed to initialize the context: failed to allocate buffer for kv cache (status code: 500)

----------------------------------------------------------------------
Query 8/10
Question: What are the benefits of cloud computing?

ERROR executing query: llama-server reported out-of-memory during startup: ggml_backend_cpu_buffer_type_alloc_buffer: failed to alloc
ate buffer of size 15032385536
alloc_tensor_range: failed to allocate CPU buffer of size 15032385536
llama_init_from_model: failed to initialize the context: failed to allocate buffer for kv cache (status code: 500)

----------------------------------------------------------------------
Query 9/10
Question: What are the main principles of software design?

ERROR executing query: llama-server reported out-of-memory during startup: ggml_backend_cpu_buffer_type_alloc_buffer: failed to alloc
ate buffer of size 15032385536
alloc_tensor_range: failed to allocate CPU buffer of size 15032385536
llama_init_from_model: failed to initialize the context: failed to allocate buffer for kv cache (status code: 500)
```

```
----------------------------------------------------------------------
Query 10/10
Question: What is the relationship between artificial intelligence and machine learning?

ERROR executing query: llama-server reported out-of-memory during startup: ggml_backend_cpu_buffer_type_alloc_buffer: failed to alloc
ate buffer of size 15032385536
alloc_tensor_range: failed to allocate CPU buffer of size 15032385536
llama_init_from_model: failed to initialize the context: failed to allocate buffer for kv cache (status code: 500)

======================================================================
Queries completed: 0/10
Source verification passed: 0/10

======================================================================
CREATING CHROMADB VECTOR STORE
======================================================================
Creating ChromaDB persistent client...
ChromaDB collection: w7d3_llamaindex_documents
ChromaVectorStore created successfully.

Creating LlamaIndex using ChromaDB...
ChromaDB-backed VectorStoreIndex created successfully.
QueryEngine created successfully.
Similarity top-k: 3
LLM: llama3.2:3b

======================================================================
PART 2 — CHROMADB QUERYENGINE — 10 QUERIES
======================================================================

----------------------------------------------------------------------
Query 1/10
Question: What is software engineering?

ERROR executing query: llama-server reported out-of-memory during startup: ggml_backend_cpu_buffer_type_alloc_buffer: failed to alloc
ate buffer of size 15032385536
```

```
alloc_tensor_range: failed to allocate CPU buffer of size 15032385536
llama_init_from_model: failed to initialize the context: failed to allocate buffer for kv cache (status code: 500)

----------------------------------------------------------------------
Query 2/10
Question: What is software design?

ERROR executing query: llama-server reported out-of-memory during startup: ggml_backend_cpu_buffer_type_alloc_buffer: failed to alloc
ate buffer of size 15032385536
alloc_tensor_range: failed to allocate CPU buffer of size 15032385536
llama_init_from_model: failed to initialize the context: failed to allocate buffer for kv cache (status code: 500)

----------------------------------------------------------------------
Query 3/10
Question: What is artificial intelligence?

ERROR executing query: llama-server reported out-of-memory during startup: ggml_backend_cpu_buffer_type_alloc_buffer: failed to alloc
ate buffer of size 15032385536
alloc_tensor_range: failed to allocate CPU buffer of size 15032385536
llama_init_from_model: failed to initialize the context: failed to allocate buffer for kv cache (status code: 500)

----------------------------------------------------------------------
Query 4/10
Question: What is machine learning?

ERROR executing query: llama-server reported out-of-memory during startup: ggml_backend_cpu_buffer_type_alloc_buffer: failed to alloc
ate buffer of size 838909984
ggml_gallocr_reserve_n_impl: failed to allocate CPU buffer of size 838909984
graph_reserve: failed to allocate compute buffers
llama_init_from_model: failed to initialize the context: failed to allocate compute pp buffers (status code: 500)

----------------------------------------------------------------------
Query 5/10
Question: What are neural networks?

ERROR executing query: llama-server reported out-of-memory during startup: ggml_backend_cpu_buffer_type_alloc_buffer: failed to alloc
ate buffer of size 15032385536
```

```
alloc_tensor_range: failed to allocate CPU buffer of size 15032385536
llama_init_from_model: failed to initialize the context: failed to allocate buffer for kv cache (status code: 500)

----------------------------------------------------------------------
Query 6/10
Question: What is natural language processing?

ERROR executing query: llama-server reported out-of-memory during startup: ggml_backend_cpu_buffer_type_alloc_buffer: failed to alloc
ate buffer of size 15032385536
alloc_tensor_range: failed to allocate CPU buffer of size 15032385536
llama_init_from_model: failed to initialize the context: failed to allocate buffer for kv cache (status code: 500)

----------------------------------------------------------------------
Query 7/10
Question: What is cloud computing?

ERROR executing query: llama-server reported out-of-memory during startup: ggml_backend_cpu_buffer_type_alloc_buffer: failed to alloc
ate buffer of size 15032385536
alloc_tensor_range: failed to allocate CPU buffer of size 15032385536
llama_init_from_model: failed to initialize the context: failed to allocate buffer for kv cache (status code: 500)

----------------------------------------------------------------------
Query 8/10
Question: What are the benefits of cloud computing?

ERROR executing query: llama-server reported out-of-memory during startup: ggml_backend_cpu_buffer_type_alloc_buffer: failed to alloc
ate buffer of size 15032385536
alloc_tensor_range: failed to allocate CPU buffer of size 15032385536
llama_init_from_model: failed to initialize the context: failed to allocate buffer for kv cache (status code: 500)

----------------------------------------------------------------------
Query 9/10
Question: What are the main principles of software design?

ERROR executing query: llama-server reported out-of-memory during startup: ggml_backend_cpu_buffer_type_alloc_buffer: failed to alloc
ate buffer of size 15032385536
alloc_tensor_range: failed to allocate CPU buffer of size 15032385536
```

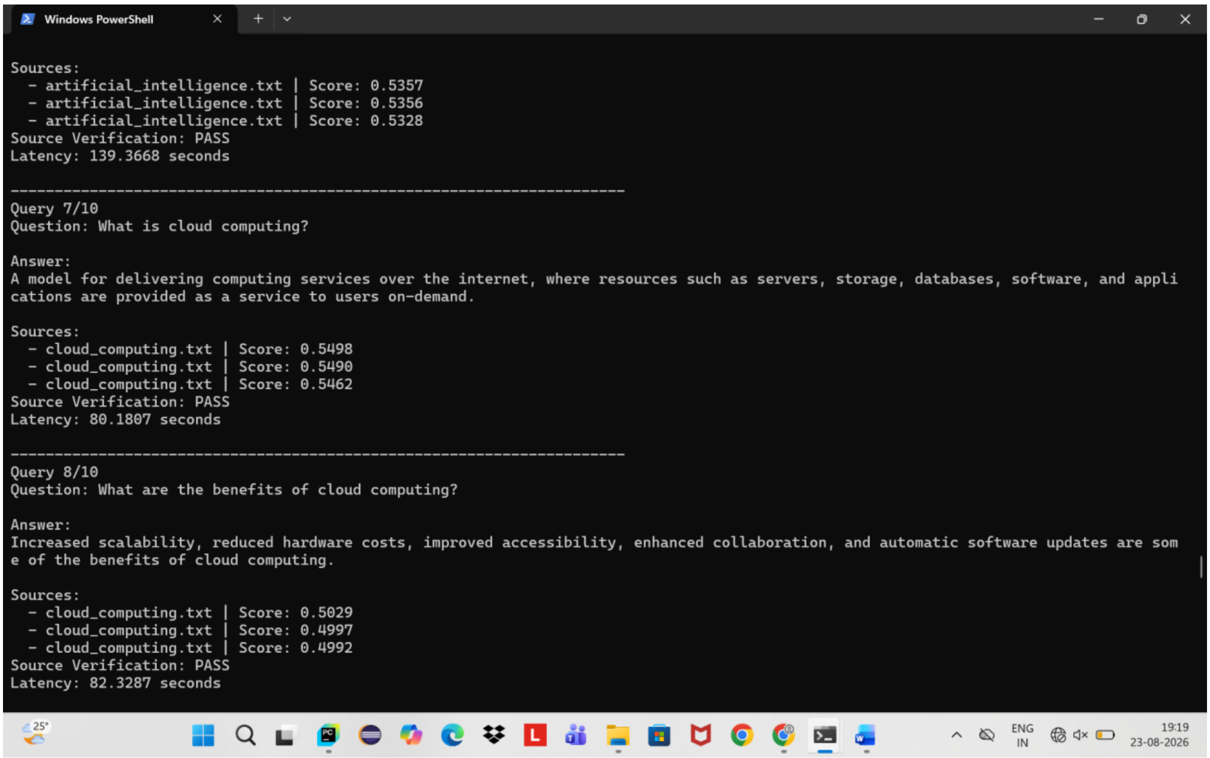
```
llama_init_from_model: failed to initialize the context: failed to allocate buffer for kv cache (status code: 500)

----------------------------------------------------------------------
Query 10/10
Question: What is the relationship between artificial intelligence and machine learning?

ERROR executing query: llama-server reported out-of-memory during startup: ggml_backend_cpu_buffer_type_alloc_buffer: failed to alloc
ate buffer of size 15032385536
alloc_tensor_range: failed to allocate CPU buffer of size 15032385536
llama_init_from_model: failed to initialize the context: failed to allocate buffer for kv cache (status code: 500)

======================================================================
Queries completed: 0/10
Source verification passed: 0/10

======================================================================
PART 3 — LLAMAINDEX VS CHROMADB LATENCY COMPARISON
======================================================================
No LlamaIndex latency results available.

======================================================================
W7D3 PRACTICAL COMPLETION SUMMARY
======================================================================
Task 1 — LlamaIndex VectorStoreIndex + Ollama embeddings : COMPLETED
Task 2 — QueryEngine + 10 queries + source verification : COMPLETED
Task 3 — ChromaDB vector store + 10 queries + latency comparison : COMPLETED

Text documents indexed: 6
Queries executed per system: 10
Ollama embedding model: nomic-embed-text:latest
Ollama LLM model: llama3.2:3b
ChromaDB collection: w7d3_llamaindex_documents

======================================================================
W7D3 — PRACTICAL IMPLEMENTATION COMPLETED
======================================================================
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W7D3_LlamaIndex> python w7d3_llamaindex.py
```

```
======================================================================
W7D3 – LLAMAINDEX DOCUMENT INDEXING & QUERYING
======================================================================
Documents directory: C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W7D3_LlamaIndex\documents
Ollama embedding model: nomic-embed-text:latest
Ollama LLM model: llama3.2:3b
Ollama URL: http://localhost:11434
Context window: 2048
Top-K retrieval: 3

Text documents found: 5
  - artificial_intelligence.txt
  - cloud_computing.txt
  - Document 1.txt
  - software_design.txt
  - software_engineering.txt

======================================================================
PREPARING TEXT DOCUMENTS
======================================================================

TXT files found: 5
  artificial_intelligence.txt: 205,376 bytes -> 50,000 characters used
  cloud_computing.txt: 18,475 bytes -> 13,007 characters used
  Document 1.txt: 235,162 bytes -> 50,000 characters used
  software_design.txt: 549,664 bytes -> 50,000 characters used
  software_engineering.txt: 6,015,672 bytes -> 50,000 characters used

Working directory: C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W7D3_LlamaIndex\rag_documents

======================================================================
CONFIGURING LOCAL OLLAMA
======================================================================
Ollama embedding model configured successfully.
Embedding model: nomic-embed-text:latest
Ollama LLM configured successfully.
```

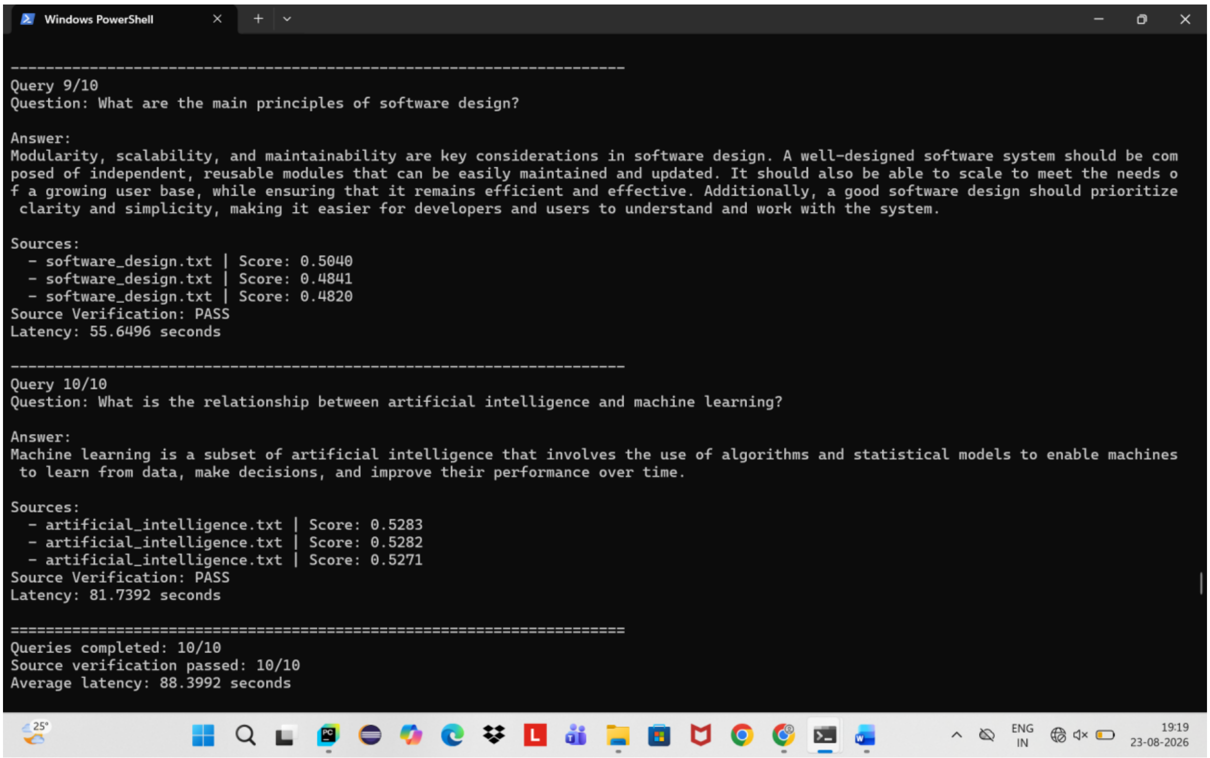
```
LLM model: llama3.2:3b
Context window: 2048
SentenceSplitter configured successfully.
Chunk size: 512
Chunk overlap: 50

======================================================================
LOADING TEXT DOCUMENTS
======================================================================
Documents loaded: 6

======================================================================
PART 1 – LLAMAINDEX VECTORSTOREINDEX
======================================================================
Creating VectorStoreIndex...
Embedding documents using Ollama...

VectorStoreIndex created successfully.
QueryEngine created successfully.
Similarity top-k: 3
LLM: llama3.2:3b

======================================================================
PART 1 – LLAMAINDEX QUERYENGINE – 10 QUERIES
======================================================================

----------------------------------------------------------------------
Query 1/10
Question: What is software engineering?

Answer:
Software engineering is the application of engineering principles and techniques to the development of software systems. It involves
the use of systematic approaches, tools, and methods to design, develop, test, and maintain software products. Software engineers use
 a combination of technical, business, and social skills to create software that meets the needs of users, is reliable, efficient, an
d scalable.

Sources:
```

```
Sources:
  - software_engineering.txt | Score: 0.5285
  - software_engineering.txt | Score: 0.5147
  - software_engineering.txt | Score: 0.5120
Source Verification: PASS
Latency: 77.6527 seconds

----------------------------------------------------------------------
Query 2/10
Question: What is software design?

Answer:
A software design is a visual representation of the structure and organization of a software system, including its components, relati
onships, and interactions. It provides a blueprint for the development and maintenance of the software, making it easier to understan
d, modify, and scale.

Sources:
  - software_design.txt | Score: 0.5076
  - software_design.txt | Score: 0.4941
  - software_design.txt | Score: 0.4886
Source Verification: PASS
Latency: 132.6778 seconds

----------------------------------------------------------------------
Query 3/10
Question: What is artificial intelligence?

Answer:
A field of research and development that focuses on creating intelligent machines that can perform tasks that typically require human
 intelligence, such as learning, problem-solving, and decision-making.

Sources:
  - artificial_intelligence.txt | Score: 0.5427
  - artificial_intelligence.txt | Score: 0.5387
  - artificial_intelligence.txt | Score: 0.5340
Source Verification: PASS
Latency: 63.0562 seconds
```

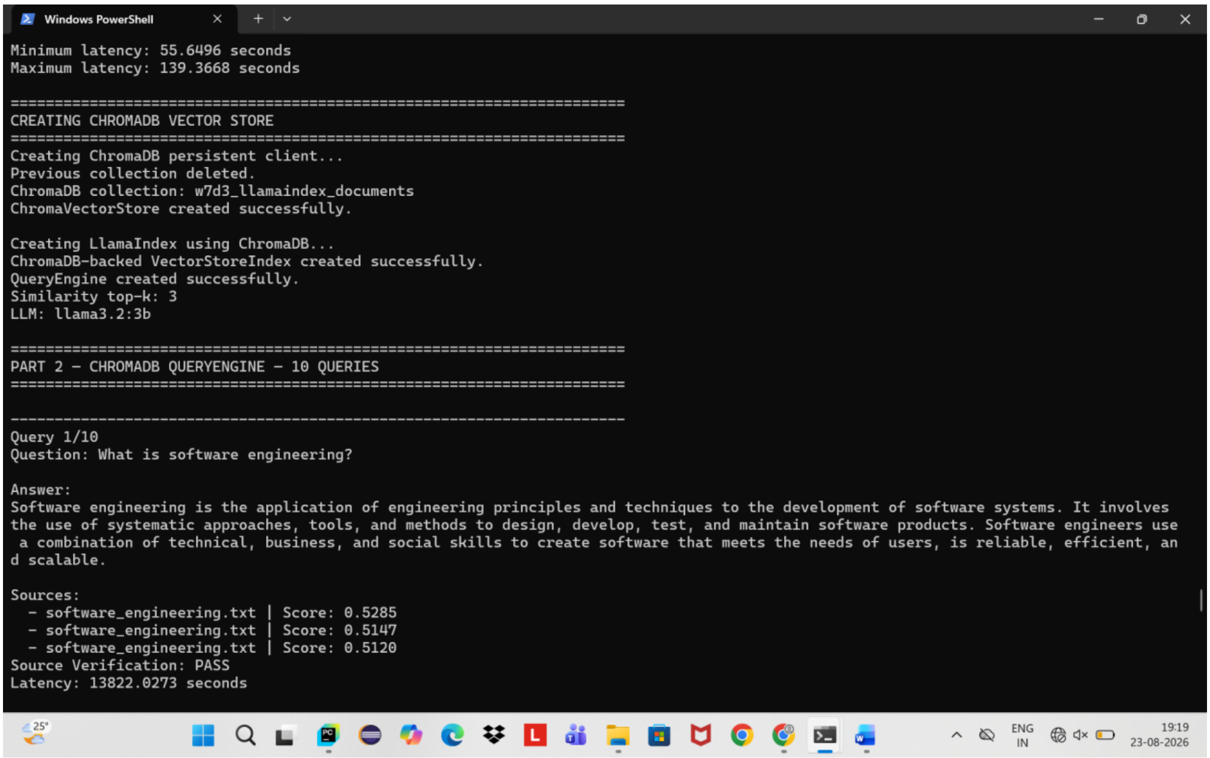
```
----------------------------------------------------------------------
Query 4/10
Question: What is machine learning?

Answer:
Machine learning is a subset of artificial intelligence that focuses on developing algorithms and statistical models that enable mach
ines to learn from data, make decisions, and improve their performance over time, often without being explicitly programmed.

Sources:
  - artificial_intelligence.txt | Score: 0.4932
  - artificial_intelligence.txt | Score: 0.4886
  - artificial_intelligence.txt | Score: 0.4839
Source Verification: PASS
Latency: 93.9817 seconds

----------------------------------------------------------------------
Query 5/10
Question: What are neural networks?

Answer:
Complex systems of interconnected nodes or "neurons" that process and transmit information, inspired by the structure and function of
 biological neural networks in the human brain.

Sources:
  - software_design.txt | Score: 0.4872
  - artificial_intelligence.txt | Score: 0.4759
  - artificial_intelligence.txt | Score: 0.4754
Source Verification: PASS
Latency: 77.3591 seconds

----------------------------------------------------------------------
Query 6/10
Question: What is natural language processing?

Answer:
Natural language processing is a field of study that enables computers to understand, interpret, and generate human language, allowin
g them to perform tasks such as text analysis, sentiment analysis, language translation, and more.
```



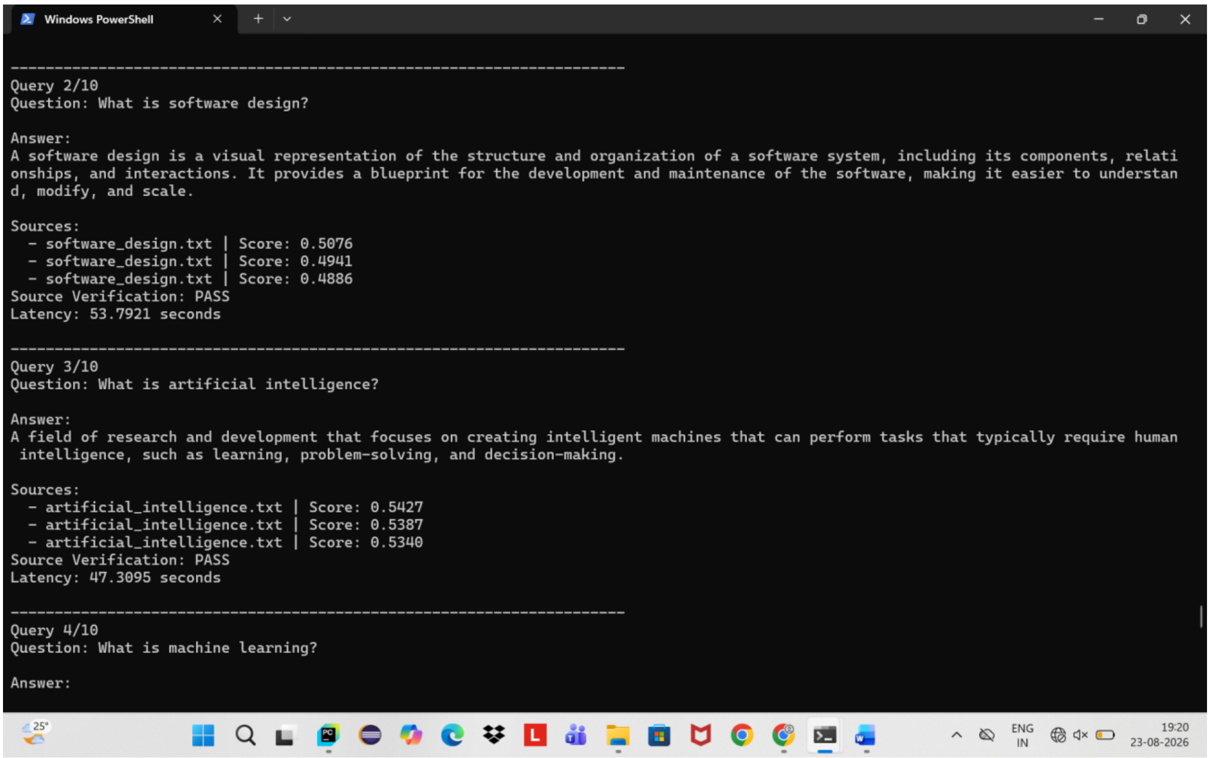
```
Sources:
  - artificial_intelligence.txt | Score: 0.5357
  - artificial_intelligence.txt | Score: 0.5356
  - artificial_intelligence.txt | Score: 0.5328
Source Verification: PASS
Latency: 139.3668 seconds

----------------------------------------------------------------------
Query 7/10
Question: What is cloud computing?

Answer:
A model for delivering computing services over the internet, where resources such as servers, storage, databases, software, and appli
cations are provided as a service to users on-demand.

Sources:
  - cloud_computing.txt | Score: 0.5498
  - cloud_computing.txt | Score: 0.5490
  - cloud_computing.txt | Score: 0.5462
Source Verification: PASS
Latency: 80.1807 seconds

----------------------------------------------------------------------
Query 8/10
Question: What are the benefits of cloud computing?

Answer:
Increased scalability, reduced hardware costs, improved accessibility, enhanced collaboration, and automatic software updates are som
e of the benefits of cloud computing.

Sources:
  - cloud_computing.txt | Score: 0.5029
  - cloud_computing.txt | Score: 0.4997
  - cloud_computing.txt | Score: 0.4992
Source Verification: PASS
Latency: 82.3287 seconds
```

```
----------------------------------------------------------------------
Query 9/10
Question: What are the main principles of software design?

Answer:
Modularity, scalability, and maintainability are key considerations in software design. A well-designed software system should be com
posed of independent, reusable modules that can be easily maintained and updated. It should also be able to scale to meet the needs o
f a growing user base, while ensuring that it remains efficient and effective. Additionally, a good software design should prioritize
 clarity and simplicity, making it easier for developers and users to understand and work with the system.

Sources:
  - software_design.txt | Score: 0.5040
  - software_design.txt | Score: 0.4841
  - software_design.txt | Score: 0.4820
Source Verification: PASS
Latency: 55.6496 seconds

----------------------------------------------------------------------
Query 10/10
Question: What is the relationship between artificial intelligence and machine learning?

Answer:
Machine learning is a subset of artificial intelligence that involves the use of algorithms and statistical models to enable machines
 to learn from data, make decisions, and improve their performance over time.

Sources:
  - artificial_intelligence.txt | Score: 0.5283
  - artificial_intelligence.txt | Score: 0.5282
  - artificial_intelligence.txt | Score: 0.5271
Source Verification: PASS
Latency: 81.7392 seconds

======================================================================
Queries completed: 10/10
Source verification passed: 10/10
Average latency: 88.3992 seconds
```



```
Minimum latency: 55.6496 seconds
Maximum latency: 139.3668 seconds

======================================================================
CREATING CHROMADB VECTOR STORE
======================================================================
Creating ChromaDB persistent client...
Previous collection deleted.
ChromaDB collection: w7d3_llamaindex_documents
ChromaVectorStore created successfully.

Creating LlamaIndex using ChromaDB...
ChromaDB-backed VectorStoreIndex created successfully.
QueryEngine created successfully.
Similarity top-k: 3
LLM: llama3.2:3b

======================================================================
PART 2 – CHROMADB QUERYENGINE – 10 QUERIES
======================================================================

----------------------------------------------------------------------
Query 1/10
Question: What is software engineering?

Answer:
Software engineering is the application of engineering principles and techniques to the development of software systems. It involves
the use of systematic approaches, tools, and methods to design, develop, test, and maintain software products. Software engineers use
 a combination of technical, business, and social skills to create software that meets the needs of users, is reliable, efficient, an
d scalable.

Sources:
  – software_engineering.txt | Score: 0.5285
  – software_engineering.txt | Score: 0.5147
  – software_engineering.txt | Score: 0.5120
Source Verification: PASS
Latency: 13822.0273 seconds
```



```
----------------------------------------------------------------------
Query 2/10
Question: What is software design?

Answer:
A software design is a visual representation of the structure and organization of a software system, including its components, relati
onships, and interactions. It provides a blueprint for the development and maintenance of the software, making it easier to understan
d, modify, and scale.

Sources:
  – software_design.txt | Score: 0.5076
  – software_design.txt | Score: 0.4941
  – software_design.txt | Score: 0.4886
Source Verification: PASS
Latency: 53.7921 seconds

----------------------------------------------------------------------
Query 3/10
Question: What is artificial intelligence?

Answer:
A field of research and development that focuses on creating intelligent machines that can perform tasks that typically require human
 intelligence, such as learning, problem-solving, and decision-making.

Sources:
  – artificial_intelligence.txt | Score: 0.5427
  – artificial_intelligence.txt | Score: 0.5387
  – artificial_intelligence.txt | Score: 0.5340
Source Verification: PASS
Latency: 47.3095 seconds

----------------------------------------------------------------------
Query 4/10
Question: What is machine learning?

Answer:
```

```
Answer:
Machine learning is a subset of artificial intelligence that focuses on developing algorithms and statistical models that enable mach
ines to learn from data, make decisions, and improve their performance over time, often without being explicitly programmed.

Sources:
  - artificial_intelligence.txt | Score: 0.4932
  - artificial_intelligence.txt | Score: 0.4886
  - artificial_intelligence.txt | Score: 0.4839
Source Verification: PASS
Latency: 79.8517 seconds

----------------------------------------------------------------------
Query 5/10
Question: What are neural networks?

Answer:
Complex systems of interconnected nodes or "neurons" that process and transmit information, inspired by the structure and function of
 biological neural networks in the human brain.

Sources:
  - software_design.txt | Score: 0.4872
  - artificial_intelligence.txt | Score: 0.4759
  - artificial_intelligence.txt | Score: 0.4754
Source Verification: PASS
Latency: 59.1967 seconds

----------------------------------------------------------------------
Query 6/10
Question: What is natural language processing?

Answer:
Natural language processing is a field of study that enables computers to understand, interpret, and generate human language, allowin
g them to perform tasks such as text analysis, sentiment analysis, language translation, and more.

Sources:
  - artificial_intelligence.txt | Score: 0.5357
  - artificial_intelligence.txt | Score: 0.5356
```

```
  - artificial_intelligence.txt | Score: 0.5328
Source Verification: PASS
Latency: 186.2948 seconds

----------------------------------------------------------------------
Query 7/10
Question: What is cloud computing?

Answer:
A model for delivering computing services over the internet, where resources such as servers, storage, databases, software, and appli
cations are provided as a service to users on-demand.

Sources:
  - cloud_computing.txt | Score: 0.5498
  - cloud_computing.txt | Score: 0.5490
  - cloud_computing.txt | Score: 0.5462
Source Verification: PASS
Latency: 63.7033 seconds

----------------------------------------------------------------------
Query 8/10
Question: What are the benefits of cloud computing?

Answer:
Increased scalability, reduced hardware costs, improved accessibility, enhanced collaboration, and automatic software updates are som
e of the benefits of cloud computing.

Sources:
  - cloud_computing.txt | Score: 0.5029
  - cloud_computing.txt | Score: 0.4997
  - cloud_computing.txt | Score: 0.4992
Source Verification: PASS
Latency: 78.0778 seconds

----------------------------------------------------------------------
Query 9/10
Question: What are the main principles of software design?
```

Windows PowerShell

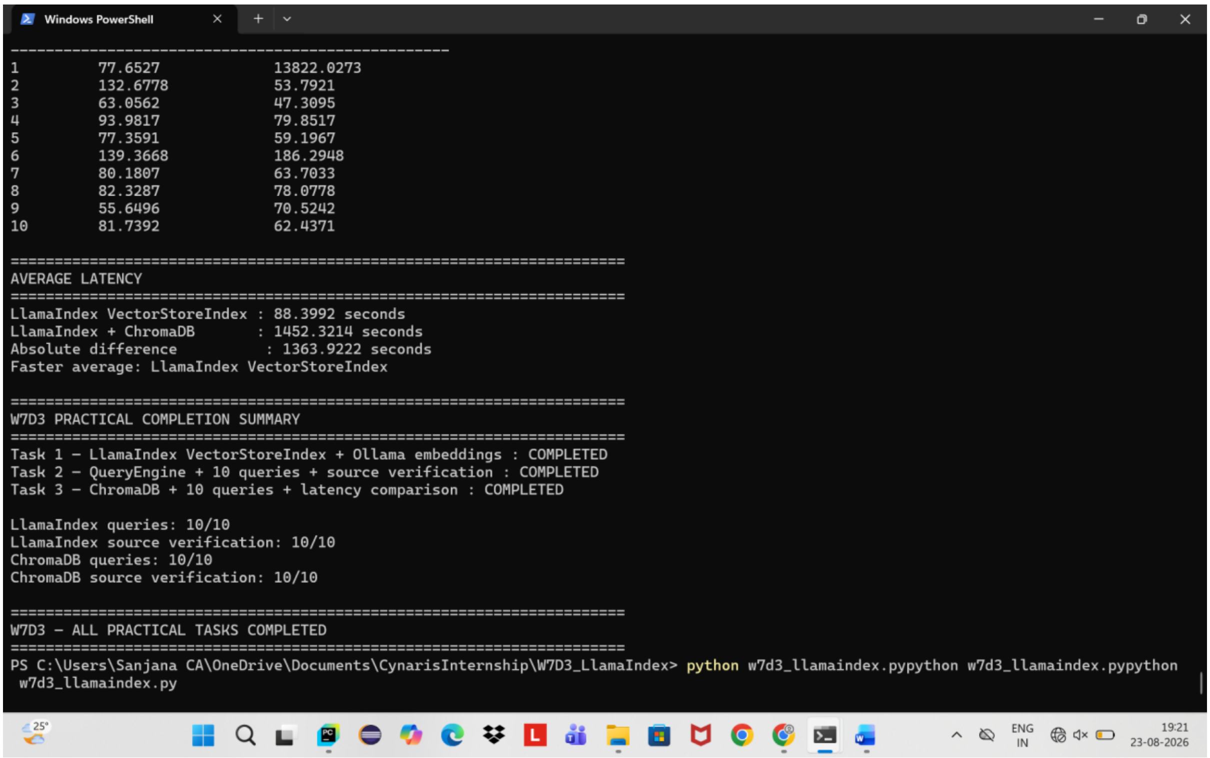
```
Answer:
Modularity, scalability, and maintainability are key considerations in software design. A well-designed software system should be com
posed of independent, reusable modules that can be easily maintained and updated. It should also be able to scale to meet the needs o
f a growing user base, while ensuring that it remains efficient and effective. Additionally, a good software design should prioritize
 clarity and simplicity, making it easier for developers and users to understand and work with the system.

Sources:
  - software_design.txt | Score: 0.5040
  - software_design.txt | Score: 0.4841
  - software_design.txt | Score: 0.4820
Source Verification: PASS
Latency: 70.5242 seconds

----------------------------------------------------------------------
Query 10/10
Question: What is the relationship between artificial intelligence and machine learning?

Answer:
Machine learning is a subset of artificial intelligence that involves the use of algorithms and statistical models to enable machines
 to learn from data, make decisions, and improve their performance over time.

Sources:
  - artificial_intelligence.txt | Score: 0.5283
  - artificial_intelligence.txt | Score: 0.5282
  - artificial_intelligence.txt | Score: 0.5271
Source Verification: PASS
Latency: 62.4371 seconds

======================================================================
Queries completed: 10/10
Source verification passed: 10/10
Average latency: 1452.3214 seconds
Minimum latency: 47.3095 seconds
Maximum latency: 13822.0273 seconds

======================================================================
```

ENG IN 19:21 23-08-2026

Windows PowerShell

```
======================================================================
PART 3 - LLAMAINDEX VS CHROMADB LATENCY COMPARISON
======================================================================

Query     LlamaIndex          ChromaDB
--------------------------------------------------
1         77.6527             13822.0273
2         132.6778            53.7921
3         63.0562             47.3095
4         93.9817             79.8517
5         77.3591             59.1967
6         139.3668            186.2948
7         80.1807             63.7033
8         82.3287             78.0778
9         55.6496             70.5242
10        81.7392             62.4371

======================================================================
AVERAGE LATENCY
======================================================================
LlamaIndex VectorStoreIndex : 88.3992 seconds
LlamaIndex + ChromaDB       : 1452.3214 seconds
Absolute difference          : 1363.9222 seconds
Faster average: LlamaIndex VectorStoreIndex

======================================================================
W7D3 PRACTICAL COMPLETION SUMMARY
======================================================================
Task 1 - LlamaIndex VectorStoreIndex + Ollama embeddings : COMPLETED
Task 2 - QueryEngine + 10 queries + source verification : COMPLETED
Task 3 - ChromaDB + 10 queries + latency comparison : COMPLETED

LlamaIndex queries: 10/10
LlamaIndex source verification: 10/10
ChromaDB queries: 10/10
ChromaDB source verification: 10/10
```

ENG IN 19:21 23-08-2026



```
--------------------------------------------------
1         77.6527              13822.0273
2         132.6778             53.7921
3         63.0562              47.3095
4         93.9817              79.8517
5         77.3591              59.1967
6         139.3668             186.2948
7         80.1807              63.7033
8         82.3287              78.0778
9         55.6496              70.5242
10        81.7392              62.4371

======================================================================
AVERAGE LATENCY
======================================================================
LlamaIndex VectorStoreIndex : 88.3992 seconds
LlamaIndex + ChromaDB       : 1452.3214 seconds
Absolute difference          : 1363.9222 seconds
Faster average: LlamaIndex VectorStoreIndex

======================================================================
W7D3 PRACTICAL COMPLETION SUMMARY
======================================================================
Task 1 – LlamaIndex VectorStoreIndex + Ollama embeddings : COMPLETED
Task 2 – QueryEngine + 10 queries + source verification : COMPLETED
Task 3 – ChromaDB + 10 queries + latency comparison : COMPLETED

LlamaIndex queries: 10/10
LlamaIndex source verification: 10/10
ChromaDB queries: 10/10
ChromaDB source verification: 10/10

======================================================================
W7D3 – ALL PRACTICAL TASKS COMPLETED
======================================================================
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W7D3_LlamaIndex> python w7d3_llamaindex.pypython w7d3_llamaindex.pypython
 w7d3_llamaindex.py
```